The empirical recurrence for OEIS A295376: recovering a conjecture that is not written down

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Abstract

OEIS A295376 records "Empirical recurrence of order 70", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 70$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 70, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 70 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295376 is "Number of nX5 0..1 arrays with each 1 horizontally or vertically adjacent to 0, 2 or 4 1s.". It has offset 1 and begins

$$13, 120, 1067, 9507, 85310, 761795, 6819658, 60996884, \ \ldots$$

The entry states:

> Empirical recurrence of order 70 (see link above)

As of the "Last modified" line on the live entry (Nov 21 2017 07:09:21, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 70$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 70$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 140$ exact terms of the model returns order 70 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{70} c_i\, a(n-i),$$

c_1	8	c_{19}	292251	c_{37}	-15139283	c_{55}	-6929
c_2	19	c_{20}	172932	c_{38}	-4828905	c_{56}	83609
c_3	-81	c_{21}	830967	c_{39}	33128424	c_{57}	-20525
c_4	-122	c_{22}	-3043843	c_{40}	31444492	c_{58}	-36588
c_5	-56	c_{23}	-312988	c_{41}	-5273739	c_{59}	-9520
c_6	1299	c_{24}	4830775	c_{42}	-23538109	c_{60}	-3109
c_7	-5053	c_{25}	7506536	c_{43}	-12103389	c_{61}	5197
c_8	2923	c_{26}	4208509	c_{44}	-455965	c_{62}	3131
c_9	12149	c_{27}	-8231548	c_{45}	614800	c_{63}	-473
c_{10}	-27886	c_{28}	-4663032	c_{46}	-3845283	c_{64}	-480
c_{11}	53752	c_{29}	14705882	c_{47}	-4050204	c_{65}	248
c_{12}	-13837	c_{30}	11688015	c_{48}	-563658	c_{66}	48
c_{13}	-7911	c_{31}	-21803129	c_{49}	3149777	c_{67}	-41
c_{14}	37596	c_{32}	-42583157	c_{50}	3183078	c_{68}	6
c_{15}	-149654	c_{33}	-28711747	c_{51}	1504968	c_{69}	-2
c_{16}	480765	c_{34}	14542851	c_{52}	86203	c_{70}	-1
c_{17}	-675141	c_{35}	33724196	c_{53}	-636896		
c_{18}	-706894	c_{36}	14736562	c_{54}	-351731		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 70, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $70+4$ terms removed returns order 70 as well, so $\deg q_{\min} = 70 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. □

4 Verification

Three checks.

First, the walk model reproduces all 18 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $70+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 70 that this sequence satisfies — and that family turns out to have exactly one member.

References

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